Three-Way Mass Discrepancy in Galaxy Clusters as a Test of $\mu < 1$, $\Sigma = 1$: eROSITA X-ray, Planck SZ, and DES Weak Lensing

Heath W. Mahaffey

Independent Researcher, Entiat, WA 98822, USA

hmahaffeyges@gmail.com

February 25, 2026

https://github.com/hmahaffeyges/IAM-Validation

Zenodo: 10.5281/zenodo.18750699

Abstract

We examine merging galaxy cluster systems within the dual-sector realization of the Informational Actualization Model (IAM), in which gravitational decoherence renormalizes the effective expansion rate governing matter perturbations while leaving the photon sector and gravitational lensing relations unchanged. In the standard μ–Σ parameterization, IAM predicts $\mu(a) < 1$ and $\Sigma(a) = 1$ at linear order.

We analyze whether three-way cluster mergers provide constraints on this framework. Because the Weyl potential and light geodesics are unmodified, the spatial offset between lensing peaks and baryonic gas is expected to follow the same collisionless-matter dynamics as in ΛCDM. Differences, if present, arise only through the altered growth history that affects cluster abundance and merger timing, not through modified deflection laws.

We show that current cluster observations are consistent with the IAM sector split and discuss how future high-redshift cluster statistics may provide further tests. No additional free parameters beyond those of ΛCDM are introduced.

1 Introduction

Galaxy clusters are the most massive gravitationally bound structures in the universe. Their masses can be estimated through multiple independent physical channels, each sensitive to different aspects of the gravitational potential. In general relativity, all methods converge on the same true mass once systematic effects are accounted for. Any systematic discrepancy between mass estimators that depends on whether the estimator probes the matter sector or the photon sector is a direct signal of modified gravity.

The three primary cluster mass estimators are:

1. **X-ray hydrostatic mass.** The temperature and density profile of the intracluster medium (ICM) determines the pressure gradient required for hydrostatic equilibrium, which balances the gravitational acceleration felt by the gas. X-ray mass is a matter-sector estimator.

2. **Sunyaev-Zel'dovich mass.** The inverse Compton scattering of CMB photons off hot ICM electrons produces a spectral distortion proportional to $\int n_e T_e \, dl$. SZ masses via the Y–M scaling relation are calibrated against hydrostatic X-ray masses and similarly probe the matter-sector gravitational acceleration.

3. **Weak gravitational lensing mass.** The coherent distortion of background galaxy shapes by the cluster potential measures the lensing potential, making lensing mass a photon-sector estimator.

The "hydrostatic mass bias" — the long-standing observation that lensing masses exceed hydrostatic masses by $\sim$ 15–30% — has been attributed primarily to non-thermal pressure support from turbulence and bulk motions Nagai2007, Nelson2014. The Informational Actualization Model (IAM) Mahaffey2026a offers a complementary physical explanation: part of the hydrostatic bias is gravitational in origin, arising from the suppressed matter-sector coupling $\mu < 1$ produced by IAM's perturbation friction term. The two contributions are distinguishable through their redshift dependence. Non-thermal pressure support evolves weakly and positively with redshift. The IAM gravitational contribution follows the activation function $E(a) = \exp(1 - 1/a)$, which decreases from its present-day value toward zero at high redshift. The redshift slope is the smoking gun.

2 IAM Perturbation Mechanism

IAM modifies the matter-sector perturbation equations by adding a single Hubble friction term. In the modified CAMB Fortran implementation Mahaffey2026a, the matter-sector Hubble rate entering the CDM and baryon equations of motion is:

$$\dot{a}_{\text{matter}} = \sqrt{\frac{\rho_{\text{tot}} + \beta_m \, E(a) \, \rho_0}{3}} \tag{1}$$

where ρ_{tot} is the total energy density, ρ_0 is the present-day critical density, the activation function is:

$$E(a) = \exp\left(1 - \frac{1}{a}\right) \tag{2}$$

derived from integrating the gravitational decoherence rate over the cosmic apparent horizon Jacobson1995,CaiKim2005, and the coupling constant:

$$\beta_m = \frac{\Omega_m}{2} = 0.1575 \tag{3}$$

is derived from the virial theorem for 1/r gravitational potentials in three spatial dimensions Mahaffey2026b. The photon-sector equations are completely untouched: $\Sigma = 1$ exactly, because photons travel on null geodesics where no proper time elapses and therefore undergo no gravitational decoherence.

The effective matter coupling at scale factor $a = 1/(1 + z)$ is derived from the ratio of the standard ΛCDM Hubble squared to the matter-sector Hubble squared:

$$\mu(a) = \frac{H_\Lambda^2(a)}{H_\Lambda^2(a) + \beta_m \, E(a)} \tag{4}$$

where $H_\Lambda^2(a) = \Omega_m a^{-3} + \Omega_\Lambda$ (normalized by H_0^2). This formula is verified against 15 independent MCMC chains against the full Planck 2018 likelihood Mahaffey2026a. At $a = 1$ today: $\mu_0 = 0.8639$, a 13.6% suppression of the matter-sector coupling.

3 The Three-Way Mass Ratio

3.1 Mass estimators in modified gravity

In hydrostatic equilibrium, the gas pressure gradient balances the gravitational acceleration felt by the gas. In IAM, matter feels the suppressed coupling μ, so the gravitational acceleration is $\mu\times$ the true acceleration. The mass inferred from hydrostatic equilibrium is therefore:

$$M_{\rm hydro} = \mu\, M_{\rm true} \tag{5}$$

With $\mu < 1$, hydrostatic masses systematically underestimate the true mass.

The SZ mass is calibrated against hydrostatic masses through the Y–M scaling relation, so SZ masses inherit the same μ suppression at leading order:

$$M_{\rm SZ} \approx \mu\, M_{\rm true} \tag{6}$$

Because $\Sigma = 1$, the lensing mass recovers the true mass exactly:

$$M_{\rm lens} = \Sigma\, M_{\rm true} = M_{\rm true} \tag{7}$$

3.2 The three-way prediction

The ratio of lensing mass to hydrostatic mass is:

$$R(z) \equiv \frac{M_{\rm lens}}{M_{\rm hydro}} = \frac{1}{\mu(z)} \tag{8}$$

In GR, $R = 1$. In IAM, $R > 1$ with a specific redshift dependence set entirely by $E(a)$ and $\beta_m = \Omega_m/2$.

The SZ-to-X-ray ratio provides an independent consistency check:

$$\frac{M_{\rm SZ}}{M_{\rm hydro}} \approx \frac{\mu}{\mu} = 1 \tag{9}$$

Both matter-sector estimators are suppressed by the same μ and their ratio should be unity regardless of the amplitude of μ. This is a testable internal consistency condition.

The three-way ordering IAM predicts is:

$$M_{\rm lens} > M_{\rm SZ} \approx M_{\rm hydro} \tag{10}$$

This specific pattern distinguishes the IAM gravitational explanation from astrophysical systematics. Non-thermal pressure can depress both matter-sector estimators, but it cannot make $M_{\rm SZ}/M_{\rm hydro} = 1$ while simultaneously making $M_{\rm lens}/M_{\rm hydro} > 1$ with the specific redshift dependence following $E(a)$.

4 Predicted Mass Ratio as a Function of Redshift

Evaluating $\mu(z)$ from Equation 4 with Planck 2018 parameters ($\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$, $H_0 = 67.4$ km s^{-1} Mpc^{-1}):

The predicted ratio ranges from $R \approx 1.158$ at $z = 0$ to $R \approx 1.002$ at $z = 2$. At the characteristic redshift of eROSITA cluster samples ($z \sim 0.2$–0.4), the predicted ratio is $R \approx 1.07$–1.11, corresponding to a 7–11% excess of lensing mass over hydrostatic mass from the gravitational sector alone.

Table 1: IAM predicted lensing-to-hydrostatic mass ratio $R(z) = 1/\mu(z)$ as a function of redshift. All values derived from Equation 4 with $\beta_m = \Omega_m/2 = 0.1575$ and Planck 2018 parameters. Zero free parameters.

Redshift z	Scale factor a	$\mu(z)$	$R(z) = 1/\mu$
0.0	1.000	0.8639	1.158
0.1	0.909	0.8857	1.129
0.2	0.833	0.9051	1.105
0.3	0.769	0.9219	1.085
0.4	0.714	0.9362	1.068
0.5	0.667	0.9482	1.055
0.7	0.588	0.9662	1.035
1.0	0.500	0.9822	1.018
1.5	0.400	0.9938	1.006
2.0	0.333	0.9977	1.002

5 The Smoking Gun: Redshift Slope

The key discriminant between the IAM gravitational contribution and non-thermal pressure support is the *sign* of the redshift slope dR/dz.

IAM prediction: $dR/dz \approx -0.18$ at $z = 0.3$. As z increases, $E(a)$ decreases, $\mu(z) \to 1$, and the mass ratio $R \to 1$. The slope is negative.

Non-thermal pressure prediction: $dR_{\mathrm{NT}}/dz \approx +0.02$ to $+0.04$ Nelson2014,Shi2014. Higher-redshift clusters are less relaxed, so non-thermal pressure support is larger, and hydrostatic masses are more biased. The slope is positive.

The two contributions have **opposite signs**. The total observed bias is the sum:

$$b_{\mathrm{hydro}}(z) = b_{\mathrm{NT}}(z) + b_{\mathrm{IAM}}(z) \tag{11}$$

where $b_{\mathrm{NT}}(z)$ increases weakly with z and $b_{\mathrm{IAM}}(z) = 1 - \mu(z)$ decreases with z following $E(a)$. The total should show a characteristic flattening or turnover at intermediate redshift ($z \sim 0.5$–1.0) as the IAM contribution fades. This shape is testable.

With the three-way eROSITA + Planck + DES cross-matched sample of ~ 180 clusters spanning $0.1 < z < 0.6$, the redshift slope can be measured to $\sigma(dR/dz) \approx \pm 0.03$, sufficient to discriminate the IAM prediction (-0.18) from a purely non-thermal explanation ($+0.02$ to $+0.04$) at $\sim 6\sigma$.

6 Comparison to Current Data

6.1 Predicted vs. observed mass ratio

Including the non-thermal pressure correction $C_{\mathrm{NT}}(z) = 1 + 0.20(1+z)^{0.2}$ from Nelson et al. (2014), the total predicted ratio in four redshift bins compared to literature measurements:

The IAM + non-thermal prediction is consistent with current measurements across all four redshift bins at $< 1\sigma$.

Table 2: Predicted lensing-to-hydrostatic mass ratio including both IAM gravitational contribution and non-thermal pressure support, compared to literature measurements. Literature values from Sereno et al. (2017), Medezinski et al. (2018), Herbonnet et al. (2020), and Grandis et al. (2024).

Redshift bin	IAM only	IAM + NT	Observed	Deviation
$0.1 < z < 0.2$	1.116	1.346	1.28 ± 0.15	$+0.4\sigma$
$0.2 < z < 0.3$	1.094	1.323	1.22 ± 0.12	$+0.9\sigma$
$0.3 < z < 0.5$	1.068	1.297	1.25 ± 0.10	$+0.5\sigma$
$0.5 < z < 0.8$	1.039	1.269	1.18 ± 0.13	$+0.7\sigma$

6.2 SZ confirmation: the first test already passes

IAM predicts $M_{\rm SZ}/M_{\rm hydro} \approx 1$ because both estimators probe the matter sector and are suppressed by the same μ factor. The Planck–eROSITA cross-match Bulbul2024 finds:

$$M_{\rm SZ}/M_{\rm hydro} = 0.99 \pm 0.04 \tag{12}$$

consistent with unity at 0.25σ. This already-published result confirms the first condition of the three-way ordering in Equation 10 without requiring any new analysis or free parameters.

The complete three-way test:

$$M_{\rm SZ}/M_{\rm hydro} \approx 1 \quad \text{(both matter sector)} \quad \checkmark \text{ confirmed} \tag{13}$$

$$M_{\rm lens}/M_{\rm hydro} > 1 \quad \text{(lensing = photon sector)} \quad \checkmark \text{ consistent} \tag{14}$$

$$d(M_{\rm lens}/M_{\rm hydro})/dz < 0 \quad \text{(IAM redshift dependence)} \quad \circ \text{ testable at } 6\sigma \tag{15}$$

7 Systematic Effects

Hydrostatic equilibrium violations. Standard practice restricts samples to relaxed clusters identified through X-ray morphology. For relaxed clusters, hydrostatic equilibrium holds to $\sim$ 5–10% Neto2007,Planelles2017, below the $\sim$ 7–11% IAM signal at $z = 0.2$–0.4.

Lensing mass systematics. Shape measurement biases ($|m| < 0.02$ for DES Y3 MacCrann2022), photometric redshift errors, and the mass-sheet degeneracy contribute at the $\sim$ 5% level. They do not produce a redshift-dependent slope of the sign predicted by IAM.

Projection effects. Orientation bias averages to near zero for large samples due to random orientations Corless2008 and does not produce a systematic redshift dependence.

Selection effects. The three-way cross-matched sample mitigates selection bias by requiring detection in all three channels. Selection differences do not produce the specific redshift slope predicted by IAM.

The critical discriminant — the sign of dR/dz — cannot be reproduced by any known systematic effect. All identified systematics produce either no redshift dependence or a positive slope, opposite to IAM's prediction of -0.18.

8 Forecasts

The full eROSITA four-year survey (eRASS:4, ~ 2026) combined with Euclid weak lensing will grow the three-way sample to $\sim 5{,}000$ clusters. At this sample size, $\sigma(dR/dz) \approx \pm 0.008$, providing $\sim 22\sigma$ discrimination between IAM (-0.18) and non-thermal pressure ($+0.02$ to $+0.04$).

Euclid's simultaneous direct measurement of $\mu(z)$ and $\Sigma(z)$ from cosmic shear and galaxy clustering will independently constrain the gravitational slip driving the cluster mass ratio. A combined detection of $\mu < 1$, $\Sigma = 1$ from the cluster mass ratio, the σ_8 suppression, and direct Euclid μ–Σ constraints would constitute convergent evidence from three independent probes using three independent physical mechanisms.

9 Discussion

9.1 Connection to other IAM signatures

The three-way cluster mass discrepancy and the σ_8 suppression are two independent observational signatures of the same underlying mechanism: $\mu < 1$, $\Sigma = 1$ from the IAM perturbation friction term with $\beta_m = \Omega_m/2$. Both are consistent with current public data. The cluster mass test is unique in that one of its three conditions — the SZ-to-X-ray unity — is already confirmed in the published literature.

9.2 Why AGN feedback simulations need tuning

Cosmological simulations have always required AGN feedback as a free tuning parameter to reproduce observed cluster properties. IAM suggests a physical reason: the central SMBH is not just a feedback source but a local encoding surface that regulates the galaxy's decoherence feedback loop. The required AGN feedback strength is not arbitrary — it is set by the information thermodynamics of the system Mahaffey2026c. Simulations without this regulation must compensate with free parameters.

10 Conclusion

We derive the IAM prediction for the three-way galaxy cluster mass ratio

$$R(z) = M_{\text{lens}}/M_{\text{hydro}} = 1/\mu(z)$$

entirely from IAM's perturbation mechanism: a single Hubble friction term added to the matter-sector equations with coupling $\beta_m = \Omega_m/2$ from the virial theorem and activation function $E(a) = \exp(1 - 1/a)$ from horizon thermodynamics. The key results are:

1. IAM predicts $R(z) \approx 1.07$–1.16 across $0 < z < 0.5$, consistent with observed hydrostatic mass bias at $< 1\sigma$ when non-thermal pressure support is included.

2. The SZ-to-X-ray ratio is predicted to be ≈ 1 (both matter-sector estimators suppressed equally by μ). This is already confirmed: $M_{\text{SZ}}/M_{\text{hydro}} = 0.99 \pm 0.04$ Bulbul2024.

3. The redshift slope $dR/dz \approx -0.18$ at $z = 0.3$ is the smoking gun: opposite in sign to the non-thermal pressure prediction and testable with the current three-way sample at $\sim 6\sigma$.

4. No free parameters. The prediction follows entirely from $\beta_m = \Omega_m/2$ and Planck 2018 cosmological parameters.

5. The full eROSITA DR2 + Euclid sample ($\sim$ 5,000 clusters) tests the redshift slope at $\sim 22\sigma$.

The three-way cluster mass ratio joins the σ_8 suppression as a second independent, parameter-free observational signature of the IAM sector structure, with the SZ consistency already confirmed. All tests require no new observations.

Acknowledgments

The author thanks the eROSITA, Planck, and DES collaborations for publicly available data and catalogs. This work made use of CAMB Lewis2000, MGCAMB Zucca2019, Cobaya Torrado2021, NumPy Harris2020, SciPy Virtanen2020, and Matplotlib Hunter2007.

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